

Tariffs, Global Value Chains, and the Incidence of Protection: Evidence from US Automobiles

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Motivation: who bears the cost of tariffs?

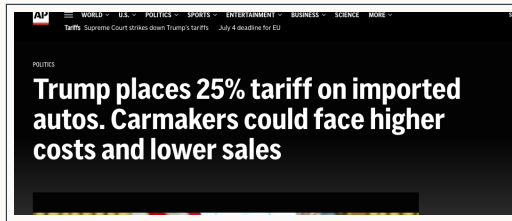
- ▶ How much will automotive tariffs affect consumer welfare? Which groups of consumers will be most affected?
- ▶ Will US firms win or lose from the tariff regime? Which firms will have positive / negative outcomes?
- ▶ How does including automotive parts in the tariff package change outcomes, and how does reliance on imported parts shape which firms win and lose?
- ▶ How do tariffs affect EV uptake? We will also model subsidy re-instatement.

Motivation: tariff incidence is in the headlines

Automakers warn that Trump tariffs will hike vehicle prices as much as 25%

By  REUTERS | [Stock Markets](#)

Published 03/04/2025, 12:34 PM | Updated 03/04/2025, 05:37 PM



AUTOS

GM lowers 2025 guidance, citing up to \$5 billion in tariff exposure

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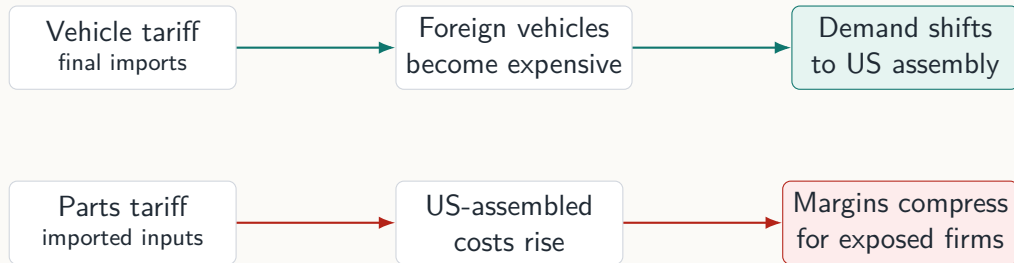
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Why Ford's made-in-America strategy hurts it in Trump's trade war

Sharon Terlep, The Wall Street Journal | 4 min read | 1 Aug 2025, 06:56 AM IST

Motivation: tariffs have opposing effects on US firms we want to disentangle



We ask how accounting for tariffs on parts changes the results of automotive tariffs.

Setting: new automotive tariffs target vehicles and components

Actual tariff regime

- ▶ Section 232 auto package: 25% additional tariffs on imported vehicles and automotive components.
- ▶ Tariff rates are lower for the UK (10%), Japan (15%), and South Korea (15%).

Setting: US automotive market is highly exposed to imports

- ▶ In the 2024 counterfactual market, 53.2% of US light-duty vehicle sales are assembled in the United States.
- ▶ Among US-assembled vehicles, 54.1% of parts value is foreign, sales-weighted.
- ▶ The included figure shows heterogeneity in foreign-parts reliance across domestically assembled vehicle lines.

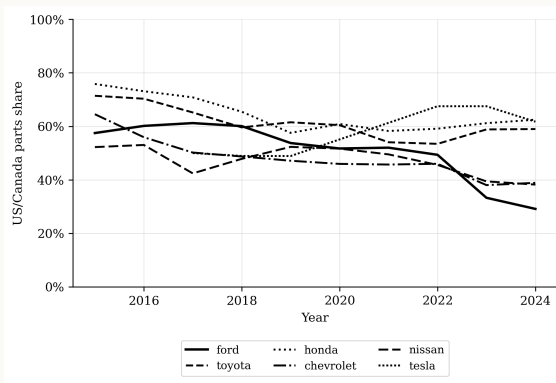


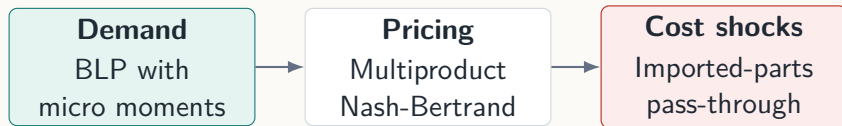
Figure reports US/Canada parts share by value. Foreign share is one minus that measure.

Results preview: parts tariffs reverse producer gains; EV subsidies boost consumer surplus

	Vehicle tariff C1	Vehicle + parts tariff C2	EV subsidy C3
Average price	+5.67%	+9.36%	−0.06%
Consumer surplus	−\$15.1 <i>b</i>	−\$31.0 <i>b</i>	+\$10.1 <i>b</i>
US producer surplus	+\$1.04 <i>b</i>	−\$2.09 <i>b</i>	+\$1.80 <i>b</i>
US assembled output	+1.25 <i>m</i>	+0.12 <i>m</i>	+0.14 <i>m</i>
EV share of sales	3.18%	3.16%	7.39%

Roadmap: how we model the impacts of tariffs

1. Build a demand system to understand how consumers substitute between products when prices change.
2. Use this demand system to recover marginal costs of each product.
3. Estimate how changes in the price of foreign parts affects the cost of domestically produced vehicles.



Data inputs

- ▶ **Sales data**

- ▶ S&P Polk registration microdata; ZIP-by-trim registrations converted to new-sales proxies and aggregated to model-year products.

- ▶ **Characteristics data**

- ▶ DataOne trim-level vehicle characteristics; median price, size, horsepower, weight, fuel economy, powertrain, and assembly location.

- ▶ **Imported parts by product**

- ▶ NHTSA/AALA manufacturing reports: assembly location, US/Canada parts share, and country-of-origin shares for foreign countries above 15% of parts value.

- ▶ **Consumer data**

- ▶ IPUMS household demographics by location and income; CEX new-vehicle expenditure moments for income-based purchase patterns.

Vehicle panel covers 2015–2024.

Model

Why use a BLP demand system?

- ▶ Tariffs move relative prices across hundreds of differentiated vehicle models; incidence depends on where consumers substitute.
- ▶ Random coefficients allow price sensitivity and tastes for size, performance, and powertrain to vary across consumers.
- ▶ The implied demand derivatives feed the supply side, recovering product-level marginal costs and markups under multiproduct Nash-Bertrand pricing.
- ▶ Micro moments discipline the key objects: demographic purchase patterns and first-second choice correlations pin down heterogeneity and substitution.

Demand identifies the substitution patterns behind incidence

$$u_{ijt} = \phi_{it} + \beta_i' X_{jt} + \alpha_{it}(p_{jt} - \tau_{jt}) + \xi_{jt} + \varepsilon_{ijt}$$

- ▶ Consumers choose among vehicles and the outside option.
- ▶ Price enters net of EV subsidy.
- ▶ Income and region shift tastes and price sensitivity.

Characteristics considered

- ▶ Price, subsidy, size, horsepower, fuel economy.
- ▶ EV, hybrid, van, truck, SUV.
- ▶ Euro brand, luxury brand, firm effects, year effects.
- ▶ Income and Census-division interactions.

Nonlinear demand estimates imply rich substitution patterns

Random coefficient standard deviations, σ

Characteristic	Coef (s.e.)
$\log(\text{mpg}_{ICE/Hyb})$	5.753 (1.049)
$\log(\text{hp})$	1.957 (0.184)
Van	6.581 (1.672)
Truck	12.456 (3.192)
SUV	2.640 (0.323)
EV	1.996 (0.987)
Euro brand	2.074 (0.392)
Luxury brand	2.707 (0.468)

Demographic and regional heterogeneity, π

Interaction	Coef (s.e.)
Intercept $\times \log(\text{income}_{10k})$	3.935 (0.931)
Price net subsidy $\times \log(\text{income}_{10k})$	4.679 (1.563)
Truck $\times$ North Central	3.512 (2.051)
Truck $\times$ South Central	3.229 (2.103)
Truck $\times$ Mountain	3.595 (2.806)
SUV $\times$ North East	0.625 (0.531)
SUV $\times$ North Central	0.553 (0.505)
EV $\times$ North East	0.893 (1.056)
EV $\times$ South Atlantic	0.693 (1.113)
EV $\times$ Mountain	1.290 (1.383)
EV $\times$ Pacific	3.204 (1.059)

Standard errors clustered by model.

Micro moments provide information on consumer substitution patterns and demographic differences

- ▶ Income moments: purchase probabilities and expected prices by income quintile.
- ▶ Geography moments: EV, truck, and SUV shares by division.
- ▶ Second-choice moments: published substitution evidence for 2015 and 2022.

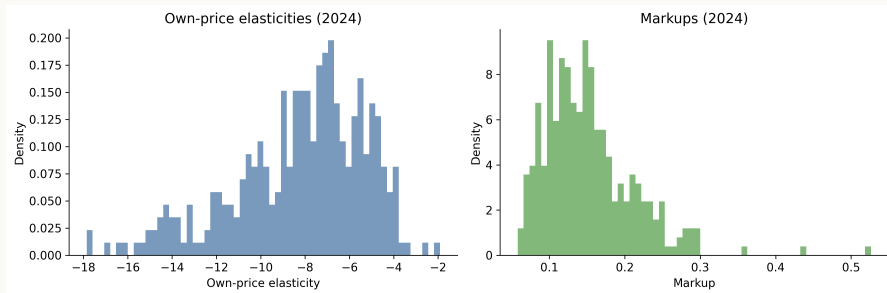
Moment	Data	Model	Diff.
Price Q5–Q1	0.063	0.071	-0.008
Purchase Q5/Q1	5.841	5.448	+0.393
2nd truck 1st truck	0.872	0.889	-0.017
2nd EV 1st EV	0.520	0.500	+0.020
Pacific EV share, 2021	0.083	0.102	-0.019
S. Central truck share, 2021	0.212	0.236	-0.024

Example diversion ratios show within-segment substitution

Product	Top destinations if buyers leave the product
Tesla Model 3 EV	Outside good 38.2%; Tesla Model Y EV 14.6%; BMW i4 EV 2.4%
Toyota RAV4 Hybrid	Outside good 30.1%; Honda CR-V Hybrid 9.3%; Hyundai Tucson Hybrid 4.1%
Ford Bronco	Outside good 18.0%; Jeep Grand Cherokee 3.9%; Chevrolet Tahoe 3.0%
Audi A5	Outside good 12.1%; Mercedes-Benz C-Class 4.9%; BMW 3 Series 2.7%
Honda Civic	Outside good 10.3%; Toyota Corolla 5.8%; Nissan Rogue 4.9%
Ford F-150	Chevrolet Silverado 29.2%; GMC Sierra 15.6%; Outside good 6.8%

- ▶ Diversion ratios are the share of consumers switching away from a product who substitute toward each alternative.
- ▶ This is the demand-side object that governs where tariff-induced price changes redirect sales.

Supply recovers marginal costs from multiproduct pricing



- Share-weighted average markup across the full sample: **17.2%**.
- Share-weighted mean own-price elasticity: **-6.764**.

We estimate the effect of foreign-parts price changes on vehicle marginal costs

$$\log(mc_{jt}) = \gamma' \log(X_{jt}) + \eta \left(\rho_{f,j,t-1}^{(1)} \log(RER_{jt}^{(1)}) \right) + \lambda_j + \lambda_t + \epsilon_{jt}$$

- ▶ $\rho_{f,j,t-1}^{(1)}$: lagged exposure to the primary foreign parts country.
- ▶ $RER_{jt}^{(1)}$: bilateral real exchange-rate shock for that country.
- ▶ Lagged exposure addresses endogeneity concerns: contemporaneous RER shocks can mechanically change reported foreign-parts value and may induce firms to reduce imported-parts shares.
- ▶ Preferred estimate: $\eta = 0.602$, meaning incomplete cost transmission.

Estimated cost-side coefficients

	Levels + controls	First differences + controls
log(size)	0.548 (0.201)	0.541 (0.254)
log(weight)	0.300 (0.176)	0.407 (0.223)
log(hp)	0.206 (0.081)	0.251 (0.098)
log(mpg)	-0.087 (0.074)	-0.046 (0.070)
$\rho_{f,j,t-1}^{(1)} \times \log(RED_{jt}^{(1)})$	0.602 (0.229)	0.513 (0.214)
Observations	384	262
R^2	0.979	0.178

Dependent variable is log(marginal cost). Standard errors clustered by make-model.

Counterfactuals re-solve prices in the 2024 market

Vehicle tariff channel

$$mc_{CF}^{(f)} = mc_0^{(f)}(1 + \tau \cdot tariff_j)$$

- ▶ Imported final vehicles receive country-specific tariff rates.

Parts tariff channel

$$mc_{CF}^{(d)} = mc_0^{(d)}(1 + \eta\rho_f \cdot tariff_{parts})$$

- ▶ US-assembled cost shock scales with imported-parts exposure.

Product offerings and assembly locations are fixed; firms re-optimize prices.

Results

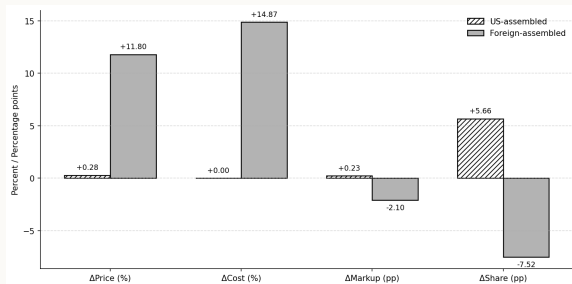
Vehicle-only tariffs look like standard protection

C1: tariff imported vehicles only

- ▶ Foreign-assembled marginal costs rise sharply.
- ▶ Domestic prices barely move.
- ▶ US-assembled market share increases by 5.66 pp.

US producer surplus rises by \$1.04b.

Consumer surplus falls by \$15.1b.



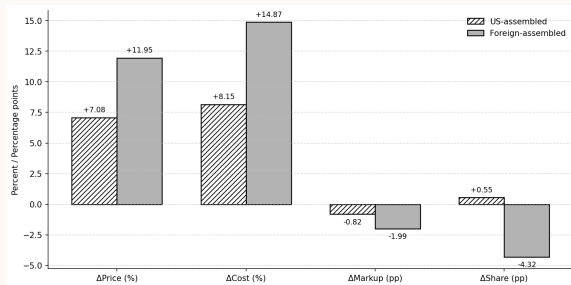
Adding parts tariffs turns protection into a broad cost shock

C2: tariff vehicles and parts

- ▶ US-assembled marginal costs rise 8.15%.
- ▶ US-assembled prices rise 7.08%.
- ▶ Domestic share gains are much smaller.

US producer surplus falls by \$2.09b.

Consumer surplus falls by \$31.0b.



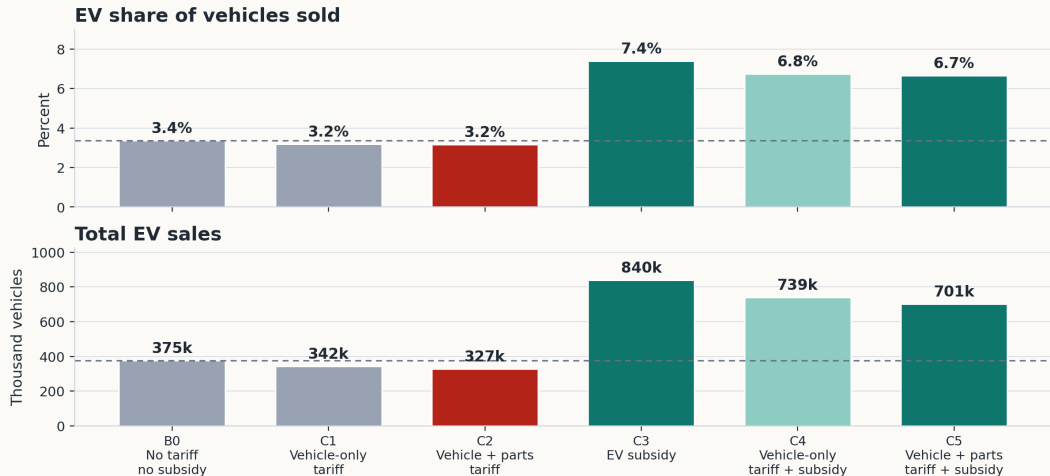
EV subsidy reenactment expands the market and EV adoption

C3: no tariff, EV subsidy

- ▶ Average prices barely move because the subsidy applies to a subset of products.
- ▶ Consumer surplus rises by \$10.1*b*.
- ▶ EV share of vehicles sold rises from 3.36% to 7.39%.

C3: EV subsidy	
Average price	-0.06%
Consumer surplus	+\$10.1 <i>b</i>
US producer surplus	+\$1.80 <i>b</i>
Vehicles sold	+0.20 <i>m</i>
EV sales	840 <i>k</i>
EV share of sales	7.39%
US assembled output	+0.14 <i>m</i>
EV subsidy cost	\$6.68 <i>b</i>

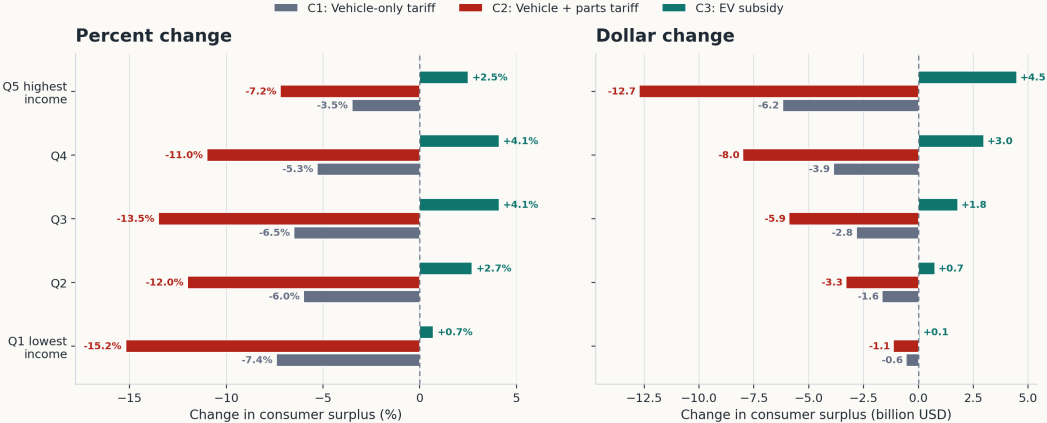
EV sales rise sharply when subsidies return



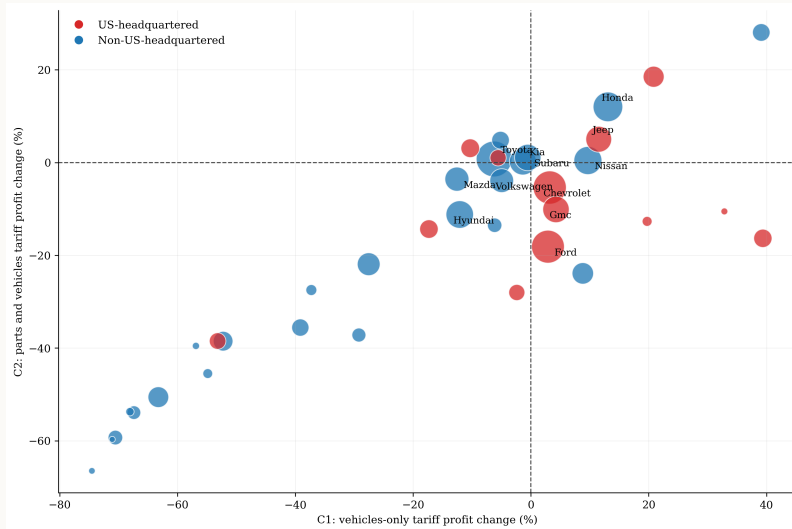
Scenario comparison

	C1: vehicles only	C2: vehicles + parts	C3: EV subsidy
Average price	+5.67%	+9.36%	−0.06%
Consumer surplus	−\$15.1 <i>b</i>	−\$31.0 <i>b</i>	+\$10.1 <i>b</i>
US producer surplus	+\$1.04 <i>b</i>	−\$2.09 <i>b</i>	+\$1.80 <i>b</i>
Vehicles sold	−0.41 <i>m</i>	−0.83 <i>m</i>	+0.20 <i>m</i>
EV share (% sales)	3.18%	3.16%	7.39%
US assembled (# vehicles)	+1.25 <i>m</i>	+0.12 <i>m</i>	+0.14 <i>m</i>
Tariff revenue	+\$15.38 <i>b</i>	+\$41.55 <i>b</i>	\$0.00 <i>b</i>
EV subsidy cost	\$0.00 <i>b</i>	\$0.00 <i>b</i>	\$6.68 <i>b</i>

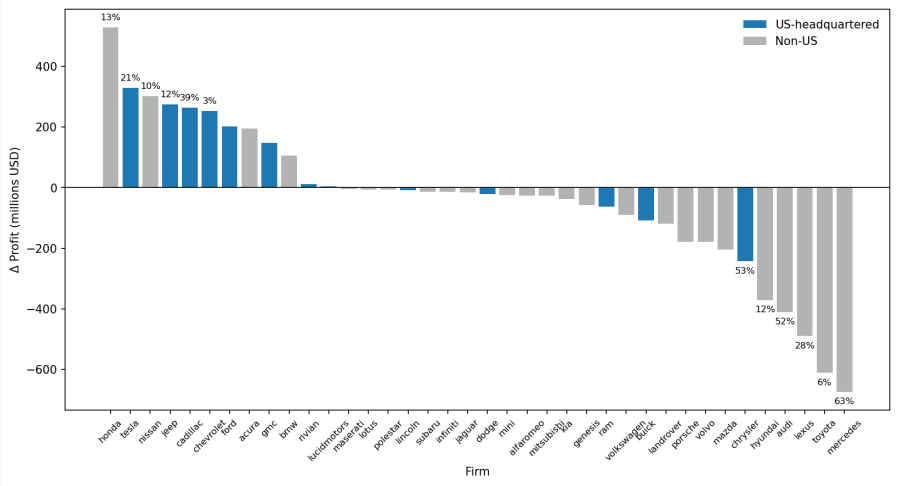
Consumer surplus impacts vary by income quintile



Firm outcomes depend on input exposure, not headquarters alone

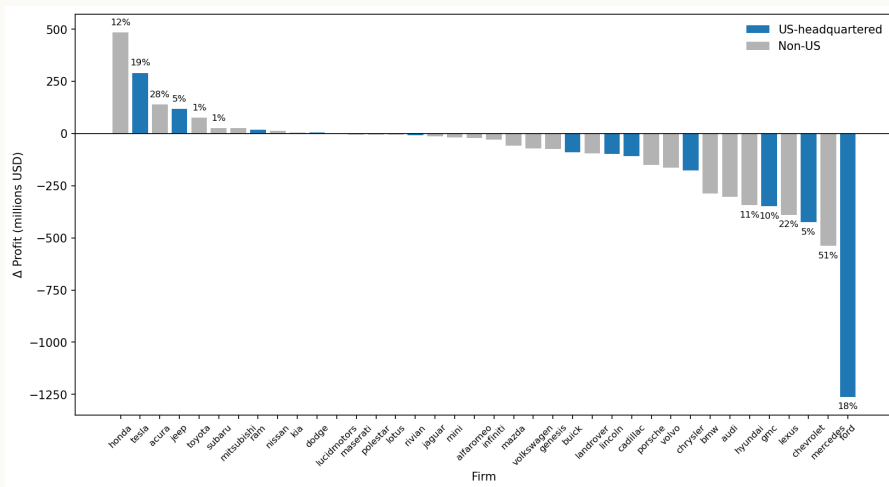


Firm-level profit changes: C1 vehicle-only tariff



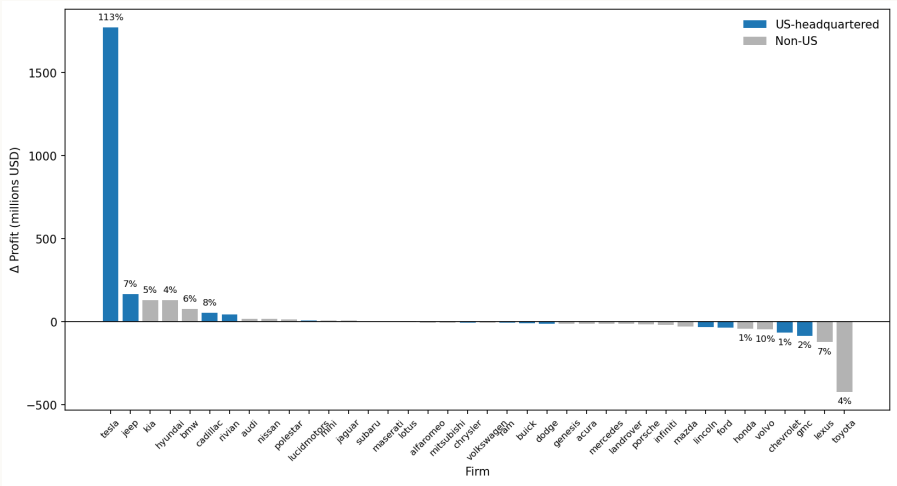
Bars report profit changes relative to B0. Blue bars are US-headquartered firms.

Firm-level profit changes: C2 vehicle + parts tariffs



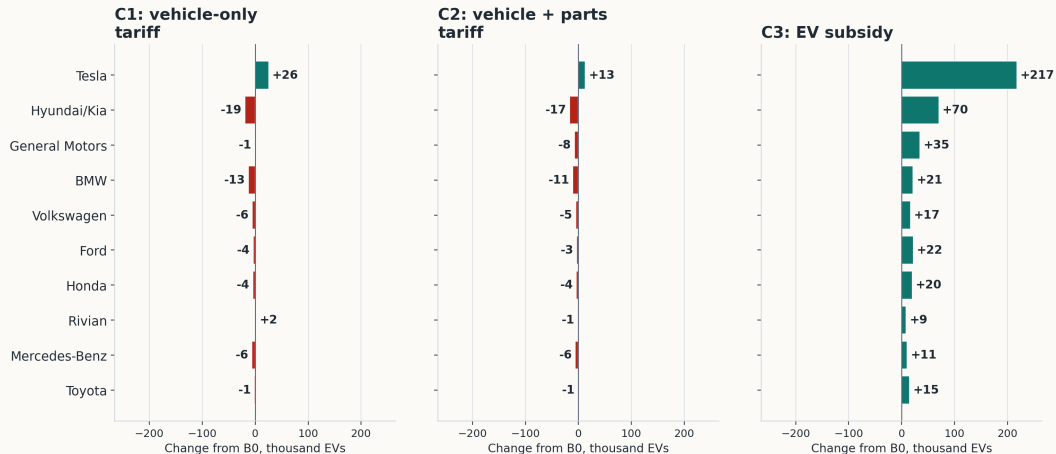
Bars report profit changes relative to B0. Blue bars are US-headquartered firms.

Firm-level profit changes: C3 EV subsidy



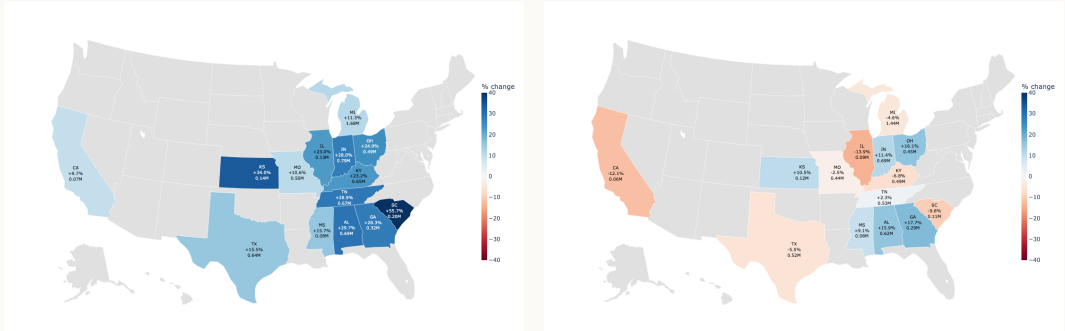
Bars report profit changes relative to B0. Blue bars are US-headquartered firms.

EV sales changes concentrate among major EV producers



Changes are relative to B0. Firms are grouped by owner.

Tariff regime choice affects production locations



Left: vehicle-only tariff. Right: vehicle and parts tariffs. Both are relative to no tariff, no subsidy.

What to take away

1. **Final-goods protection can help domestic producers.**

In this market, a vehicle-only tariff raises US producer surplus by about \$1*b*.

2. **Input tariffs can reverse that gain.**

Adding parts tariffs roughly doubles consumer losses and lowers US producer surplus by about \$2.1*b*.

3. **EV subsidy reenactment greatly expands consumer surplus and EV share.**

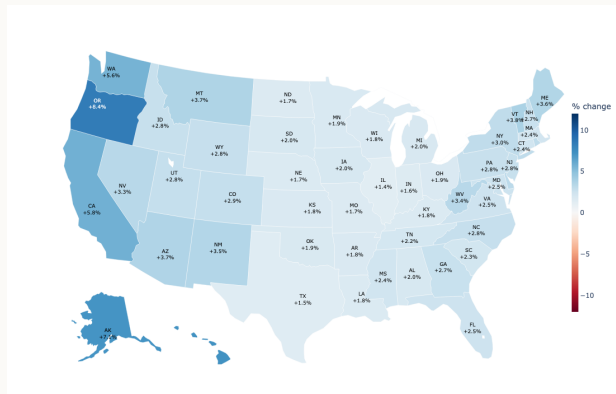
It raises consumer surplus by \$10.1*b* and increases EV share of sales from 3.36% to 7.39%.

Backup

Additional tables and robustness

Backup: regional consumer gains follow EV preferences and incomes

- ▶ Welfare gains are broad across states.
- ▶ Gains are larger in high-EV-preference regions.
- ▶ Oregon, Washington, California, Alaska, and Hawaii stand out.



Backup: what the analysis abstracts from

- ▶ Product offerings and assembly locations are fixed in the 2024 counterfactuals.
- ▶ Long-run entry, exit, plant relocation, and deep supplier-network redesign are outside the model.
- ▶ AALA data report US and Canadian parts together, so Canadian input exposure is not separately identified.
- ▶ Sourcing choices enter through the estimated pass-through sufficient statistic rather than a structural supplier-choice model.

These limits make the estimates medium-run incidence results, not long-run industrial-policy forecasts.

Backup: full six-scenario counterfactual summary

	B0	C1	C2	C3	C4	C5
Price Δ (%)	0.00	5.67	9.36	-0.06	5.61	9.30
US PS Δ (\$b)	0.00	1.04	-2.09	1.80	3.12	-0.10
CS Δ (\$b)	0.00	-15.09	-31.04	10.06	-6.19	-22.66
Vehicles sold Δ (m)	0.00	-0.41	-0.83	0.20	-0.22	-0.65
EV share (% sales)	3.36	3.18	3.16	7.39	6.75	6.67
Tariff revenue (\$b)	0.00	15.38	41.55	0.00	15.67	42.21
Subsidy cost (\$b)	0.00	0.00	0.00	6.68	6.00	5.63
Net US impact (\$b)	0.00	1.33	8.43	5.17	6.60	13.83

C1/C2 are without subsidy; C3/C4/C5 are with subsidy. C4 is vehicle-only tariff with subsidy; C5 is vehicle+parts tariff with subsidy.

Backup: distribution of consumer price sensitivity

- ▶ Estimated price disutility varies with income.
- ▶ This heterogeneity drives distributional CS results.
- ▶ Higher-income buyers account for more vehicle purchases.

